

SAENTIS, Switzerland

WMO: 066800

Lat: 47.2494N

Lon: 9.3436E

Elev: 8202

StdP: 10.83

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.5	1.9	-28.7	1.5	24.0	-22.4	2.2	23.4	51.0	25.6	46.6	24.6	13.8	270	0.936

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	58.9	49.4	56.4	47.9	54.1	46.6	51.6	55.8	49.8	54.1	48.2	52.1	6.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.8	72.2	53.4	47.9	67.2	51.3	46.3	63.3	49.9	24.4	56.0	23.1	54.2	22.1	52.2	59.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.7	36.0	30.9	DB	-4.5	64.9	5.0	7.8	-8.1	70.5	-11.1	75.1	-13.9	79.5	-17.5	85.1	(4)
(5)			WB	-5.6	55.6	4.7	2.0	-9.0	57.0	-11.7	58.2	-14.3	59.3	-17.7	60.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	19.4	18.7	22.1	26.8	34.0	40.6	43.9	44.4	38.6	34.7	26.1	21.6		
	DBStd	12.05	7.78	8.83	8.09	7.45	6.93	7.83	6.94	7.06	7.32	7.95	8.94	8.32		
	HDD50	6997	949	875	865	696	496	294	210	198	345	473	717	879		
	HDD65	12411	1414	1295	1330	1146	961	731	655	638	792	938	1167	1344		
	CDD50	60	0	0	0	0	0	12	20	25	3	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	4	0	0	0	0	0	4	0	0	0	0	0	0		
	CDH80	2	0	0	0	0	0	2	0	0	0	0	0	0		
	WSAvg	12.5	16.3	15.7	15.0	10.8	10.2	9.5	9.7	9.4	10.0	12.4	14.5	16.9		
	PrecAvg	99.1	8.6	7.4	7.9	7.4	7.8	9.4	10.1	10.5	7.5	6.2	7.4	9.3		
(15) Precipitation	PrecMax	140.1	22.6	30.4	26.5	18.0	19.8	20.9	19.3	21.2	17.7	16.3	27.5	30.7		
	PrecMin	65.9	0.2	0.7	1.0	0.5	2.0	3.3	2.0	1.9	1.1	0.5	0.0	0.6		
	PrecStd	19.6	5.6	5.3	5.6	4.1	4.1	3.5	4.1	4.4	3.5	3.9	4.9	6.9		
	PrecStd	19.6	5.6	5.3	5.6	4.1	4.1	3.5	4.1	4.4	3.5	3.9	4.9	6.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	41.0	40.7	44.7	52.2	62.0	61.6	62.2	56.0	52.6	46.2	40.4		
		MCWB	28.8	29.4	28.7	35.5	43.1	48.0	50.1	50.5	43.9	39.8	33.5	28.2		
	2%	DB	35.7	37.3	37.4	41.7	49.0	56.9	58.2	59.0	52.3	49.6	42.8	37.4		
		MCWB	26.2	26.6	28.2	34.0	41.5	47.7	49.6	49.3	42.7	38.3	32.2	26.1		
	5%	DB	32.0	33.8	34.9	39.3	46.4	53.9	55.7	56.5	50.2	47.1	40.1	34.7		
		MCWB	25.1	25.2	27.5	33.3	40.0	46.8	48.3	48.0	41.6	37.0	30.9	25.9		
	10%	DB	29.4	30.1	32.5	36.7	43.6	51.2	53.2	54.0	48.1	44.6	37.6	32.1		
		MCWB	24.2	24.3	27.2	31.5	38.5	45.4	46.5	46.9	41.6	36.2	30.5	26.1		
	0.4%	WB	32.2	33.6	33.5	38.6	46.1	52.9	54.0	54.2	48.6	43.5	37.8	32.1		
		MCDB	33.6	39.4	35.7	41.0	50.1	56.9	58.4	58.2	51.7	47.5	42.5	34.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.4	28.8	31.5	35.9	43.3	50.2	51.5	51.6	46.4	41.3	35.3	30.2		
		MCDB	32.3	32.2	34.1	39.5	46.9	54.4	55.9	55.9	49.3	45.6	38.8	32.8		
	5%	WB	27.5	26.8	29.8	34.1	41.0	48.2	49.6	49.7	44.6	39.8	33.3	28.7		
		MCDB	30.5	31.1	32.4	37.9	45.0	52.1	53.6	54.3	48.1	44.1	37.0	31.5		
	10%	WB	25.2	25.0	28.0	32.2	39.1	46.3	48.0	47.9	42.9	38.3	31.5	27.2		
		MCDB	28.5	29.9	31.0	35.7	42.5	50.2	51.7	52.1	46.4	42.6	35.6	30.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.3	8.0	7.2	7.2	7.9	8.0	7.9	7.6	7.9	7.8	8.3		
		MCDBR	9.9	12.0	9.2	7.6	8.9	10.8	10.4	10.1	9.0	9.2	9.0	9.8		
		MCWBR	6.9	8.3	6.6	7.3	8.4	10.1	9.4	9.9	8.9	7.4	5.6	6.8		
		MCDBR	10.7	11.4	8.7	7.2	8.8	9.7	9.5	9.4	8.4	8.2	8.0	9.5		
	5% WB	MCWBR	9.9	9.8	7.9	7.4	8.5	9.8	9.5	9.5	8.8	7.3	6.9	8.7		
		MCWBR	9.9	9.8	7.9	7.4	8.5	9.8	9.5	9.5	8.8	7.3	6.9	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.210	0.225	0.250	0.275	0.287	0.293	0.294	0.289	0.271	0.253	0.231	0.208		
	taud		2.496	2.474	2.506	2.522	2.531	2.537	2.521	2.552	2.600	2.638	2.619	2.519		
	Ebn at Noon		296	310	315	313	311	308	306	304	303	296	285	285		
	Edh at Noon		19	24	28	30	31	31	32	29	25	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		398	674	1058	1383	1568	1808	1761	1535	1160	739	422	328		
	RadStd		45	88	144	204	157	167	159	154	118	82	54	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+1.69	N/A	+0.61	N/A	N/A	N/A	N/A
(47)	Regional (121 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.61	N/A	N/A	+103	+27

Nomenclature: See separate page